

SAKATA, Japan

WMO: 475870

Lat: 38.9086N

Lon: 139.8440E

Elev: 5

StdP: 101.27

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-2.2	-9.3	1.7	0.1	-8.3	1.9	0.6	13.4	1.3	12.4	0.7	6.2	310	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	32.3	24.8	30.8	24.4	29.5	23.9	25.7	30.1	25.2	29.3	24.6	28.5	4.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	19.6	28.0	24.0	18.9	27.7	23.4	18.2	27.2	79.5	30.2	77.2	29.3	74.8	28.6	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	10.3	9.3	-5.2	35.1	1.4	1.6	-6.3	36.3	-7.1	37.2	-7.9	38.0	-9.0	39.2		
			-6.0	26.7	1.4	0.6	-7.0	27.1	-7.8	27.4	-8.6	27.8	-9.6	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	1.8	2.1	5.0	10.2	15.9	20.1	24.2	25.8	21.8	15.7	9.7	4.5
	DBStd	8.78	2.17	2.51	2.80	3.30	3.01	2.45	2.50	2.42	3.06	2.83	3.21	2.70
	HDD10.0	888	256	222	157	37	1	0	0	0	1	44	171	
	HDD18.3	2507	514	456	414	244	86	11	0	0	89	261	428	
	CDD10.0	2031	0	0	2	43	185	304	439	491	354	178	34	1
	CDD18.3	608	0	0	0	1	12	64	181	232	109	9	0	0
	CDH23.3	4252	0	0	0	3	63	234	1228	2093	613	17	0	0
	CDH26.7	1209	0	0	0	0	6	32	337	688	144	2	0	0
Wind	WSAvg	4.3	5.6	5.2	4.8	4.3	3.8	3.4	3.2	3.3	3.6	3.9	4.6	5.4
Precipitation	PrecAvg	1922	181	117	111	101	119	133	216	185	159	174	208	217
	PrecMax	2312	280	211	188	157	254	349	646	368	339	283	393	327
	PrecMin	1371	106	48	61	20	41	31	82	44	76	88	78	79
	PrecStd	216	42	39	34	36	56	80	112	92	69	48	69	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	11.9	17.2	23.3	27.5	29.6	33.5	34.7	32.6	25.2	20.0	14.2
		MCWB	7.2	7.5	9.9	13.8	17.3	21.0	25.0	25.2	24.3	19.1	14.7	10.9
	2%	DB	7.4	9.1	13.8	20.3	25.1	27.4	31.6	32.9	30.1	23.2	18.0	11.7
		MCWB	4.9	6.2	8.6	13.0	16.5	20.0	24.4	25.1	23.7	17.8	13.7	8.9
	5%	DB	6.2	7.4	11.6	17.8	23.2	25.8	29.9	31.5	28.2	21.9	16.2	10.2
		MCWB	3.7	4.6	7.7	11.4	15.8	19.5	24.1	24.9	22.8	17.1	12.7	7.5
	10%	DB	5.1	6.0	9.8	15.8	21.4	24.5	28.5	30.2	26.6	20.6	14.8	8.8
		MCWB	2.7	3.2	6.7	10.8	15.4	19.1	23.5	24.5	22.0	16.3	11.6	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	8.7	11.0	15.7	19.1	22.8	26.1	26.7	25.4	20.7	16.3	11.6
		MCDB	9.1	10.5	14.6	19.3	23.4	26.0	31.0	31.0	30.3	23.5	18.5	13.4
	2%	WB	5.4	6.5	9.4	13.9	18.0	21.7	25.2	26.1	24.3	19.3	14.6	9.3
		MCDB	7.0	8.7	12.5	18.5	22.3	24.7	29.5	30.4	28.5	21.8	16.9	11.3
	5%	WB	4.1	5.0	8.3	12.6	17.1	21.0	24.7	25.6	23.6	18.1	13.2	7.8
		MCDB	5.8	7.2	11.0	16.2	21.2	24.0	28.6	29.7	27.3	21.0	15.6	9.9
	10%	WB	3.0	3.5	7.0	11.6	16.2	20.2	24.1	25.1	22.7	17.0	12.0	6.3
		MCDB	4.8	5.7	9.9	15.0	20.3	23.3	27.7	29.1	26.0	19.9	14.6	8.5
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	6.7	8.1	7.9	6.7	6.2	7.0	7.2	7.6	6.9	5.2
		MCDBR	6.3	8.0	10.8	12.1	11.5	9.3	8.4	8.6	8.6	9.0	9.3	7.6
		MCWBR	5.1	6.1	7.2	6.6	5.4	4.2	3.6	3.5	4.0	5.0	6.4	6.3
	5% WB	MCDBR	6.4	7.8	9.2	9.3	8.7	6.8	7.2	7.3	7.5	7.2	8.2	7.3
		MCWBR	5.6	6.6	7.1	5.9	5.0	3.5	3.3	3.4	3.8	5.0	6.7	6.5
Clear-Sky Solar Irradiance	taub	0.408	0.443	0.509	0.546	0.556	0.540	0.551	0.523	0.463	0.435	0.426	0.417	
	taud	1.959	1.871	1.813	1.805	1.844	1.969	2.005	2.092	2.218	2.174	2.098	2.005	
	Ebn at Noon	712	745	740	745	747	760	747	758	782	756	697	668	
	Edh at Noon	133	166	196	209	205	181	174	155	129	122	116	119	
All-Sky Solar Radiation	RadAvg	1.35	2.14	3.30	4.53	5.08	5.16	4.54	4.74	3.82	2.86	1.77	1.19	
	RadStd	0.12	0.22	0.30	0.45	0.53	0.44	0.61	0.55	0.28	0.22	0.16	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)			N/A	N/A	N/A	N/A	N/A	+0.31	N/A	N/A	N/A

Nomenclature: See separate page